CDIE evidence and clarification

8 November 2026

The evidence bundle contains the requested scripts, test splits, prediction CSVs, model manifest and verification, drug metadata, and BV-BRC table. The prediction CSVs include uncertainty, adaptive threshold, live label and confidence.

Drug	n (R/S)	Sensitivity	Specificity	VME	ME	Low confidence	AUC (95% CI)
Meropenem	124 (70/54)	91.4%	87.0%	8.6% (6/70)	13.0% (7/54)	0/124	0.9421 (0.9009–0.9832)
Ciprofloxacin	143 (70/73)	80.0%	83.6%	20.0% (14/70)	16.4% (12/73)	0/143	0.8773 (0.8189–0.9357)
Cefotaxime	145 (71/74)	83.1%	74.3%	16.9% (12/71)	25.7% (19/74)	0/145	0.8723 (0.8132–0.9314)

These metrics are identical at fixed 0.50 and under the deployed adaptive rule on all 412 test rows. The live rule is `0.50 + min(0.5 × ensemble standard deviation, 0.15)`. A separate 30-isolate check against the local `predict_single_antibiotic()` path found no label or confidence mismatches and probability differences below 7.3e-7.

Model, thresholds, and calibration

Five saved `CalibratedClassifierCV` objects are loaded per drug. Each wraps a soft-voting `VotingClassifier` with XGBoost, Random Forest and LightGBM. This is five calibrated top-level models per drug, not one 15-model ensemble. Local `joblib.load()` inspection confirms the classes; Sam's Render review reports that production loads these `models_v4` assets and that all 21 deployed-file hashes match the earlier supplied manifest. The current manifest records the asset hashes; direct Render access was not available for this review.

The current cefotaxime artifact is from `models_v4`; the prediction output is named `test_predictions_cefotaxime_v4.csv`. Its reconstructed holdout AUC is 0.872288. `training_summary_v3.csv` records 0.872097 and 79.31% accuracy, but the corresponding v3 model and row-level predictions are unavailable, so that result cannot be reconciled or claimed to describe the current artifact. The current v4 CSV and `summary_cefotaxime.json` agree at 0.872288 and 78.62% (114/145).

Each deployed `metadata_{drug}.csv` has 600 NCBI labels and zero BV-BRC labels. `requirements-ml.txt` declares scikit-learn 1.7.2 and XGBoost 3.2.0; the serialized estimator confirms scikit-learn 1.7.2. The exact XGBoost training version is not established, and loading emits a serialized-model compatibility warning. Local verification used scikit-learn 1.8.0 and XGBoost 3.2.0.

The deployed code sets a 0.50 base threshold, a 0.15 maximum increase, and HIGH/MEDIUM/LOW uncertainty cutoffs of 0.05 and 0.15. The code comment says the increase is intended to reduce false positives. No documented threshold-selection or clinical decision-cost analysis was found; it can increase false-susceptible calls and is not a validated safety feature. No evidence establishes whether test results informed the original choice. All current test rows are HIGH and no calls changed under the adaptive rule.

`CalibratedClassifierCV(method="sigmoid", cv=5)` is fitted on training-partition data. The test partition is not used to fit this Platt calibrator, but it is used for reported evaluation and for the saved conformal nonconformity scores. The earlier statement that test data was not used for calibration scores was imprecise.

Previous figures and split differences

The previous 0.9871 and 0.9559 AUCs came from the `threshold_audit` split, which did not reproduce the training split and overlapped training rows. They are withdrawn as holdout results. Earlier counts of 125/140/147 also came from that non-equivalent split. The reproduced split deduplicates assembly accessions, shuffles with seed 42, excludes missing/unreadable FASTAs, then performs a stratified 25% split; it yields 124/143/145.

The earlier cefotaxime 79.31% accuracy corresponds to 115/145, while the verified v4 output is 114/145 (78.62%). The old row-level predictions are unavailable, so the one-call discrepancy cannot be traced; 79.31% is superseded.

The 0.9333 BV-BRC cefotaxime AUC is reproducible from the 500-row benchmark CSV (0.933336 from stored probabilities). At the stored 0.20 operating point the confusion counts are TP=207, TN=238, FP=12, FN=43. This is an in-corpus benchmark, not independent external validation; the rationale for 0.20 is undocumented. The 0.7793 combined-model AUC appears only in a training summary; its 1,000-row prediction output is unavailable, so we withdraw it from reported performance.

No ANI or SNP-cluster analysis has been run; relatedness across the random split is unknown. The relatedness tool is not installed in this environment, so this remains open. Test genome accessions and SHA-256 values are in the test manifest, but the FASTA files are not tracked in Git. Full sequence-level regeneration requires those inputs; the CSV verification script independently recomputes metrics without them.

The "two Kenyan isolates" figure counts unique accessions in the deployed NCBI metadata (1,673 unique accessions across drugs). A separate enrichment CSV contains seven Kenyan *K. pneumoniae* meropenem records; the PRJNA717739 attribution is from the supplied review message, not a field in that CSV. The two accessions present in current metadata are GCA_019927625.1 and GCA_020407015.1; both are in training, not the test split. The all-metadata genome manifest records the exact FASTA accession version and checksum for each required assembly. GCA_019927405.1 is not in that manifest or local `data/genomes/`, so it was not used for these results.

Repository and remaining checks

The website and backend Supabase projects will remain separate through the CDIE deadline; afterward, their boundaries should be reviewed against data trust, access control and operational ownership before deciding whether to consolidate. Backend region, live RLS, and Render key role remain unverified. The local backend migration points to the website project and defines permissive policies, while backend configuration names another project. Prediction records, feedback and IP logs have no automated retention policy; uploaded FASTA files are not retained. The project owner still needs to verify the region and key role, provide Table Editor screenshots, and grant the requested read-only access.

The requested `metadata_{drug}.csv` and `BVBRC_genome_amr.csv` are included in the repository branch. The tracked `.venv` and obsolete `data/models/` and `data/transformers/` assets are removed from the branch index without deleting local copies. The current loader uses the v4 directories. "E.coli isolates.csv" was not found in the repository; its source cannot be identified without a path or copy. `populate_isolate_table.py` was not run. The model files in branch history are ordinary Git objects; the Dockerfile has no Git LFS pull step.

Independent verification: `ALL_DRUGS_PASS: true`. Reproduction commands:

```
python scripts/regenerate_test_predictions.py --workers 8
python scripts/verify_regenerated_predictions.py
python scripts/check_live_prediction_consistency.py
```